Bharath B
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Professional Summary:

- To obtain a challenging Data Engineer position that leverages my 10+ years of experience in the software industry, including 6+ years of experience in Amazon Web services and Big data technologies, and 4+ years of experience in Data warehouse implementations.
- Results-oriented and highly skilled professional with expertise in Snowflake, AWS, and Big Data technologies.
- Proficient in using SnowSQL for complex data manipulation tasks and developing efficient data pipelines.
- Experienced in partitioning strategies and multi-cluster warehouses in Snowflake to ensure optimal query performance and scalability.
- Managed a core MarTech portfolio, made strategic decisions on integrating technology solutions.
- Hands on experience of Data Science libraries in Python such as Pandas, Numpy, SciPy, scikit-learn, Matplotlib, Seaborn, Beautiful Soup, Orange, Rpy2, LibSVM, neurolab, NLTK.
- Skilled in designing roles, views, and implementing performance tuning techniques to enhance Snowflake system performance.
- Proficient in utilizing virtual warehouses, caching, and Snowpipe for real-time data ingestion and processing in Snowflake.
- Strong knowledge of AWS CloudWatch for monitoring and managing AWS resources, setting up alarms, and collecting metrics.
- Proficient in managing user access and permissions to AWS resources using IAM.
- Extensive experience in working with HDFS, Sqoop, PySpark, Hive, MapReduce, and HBase for big data processing and analytics.
- Proficient in developing and optimizing Spark and Spark Streaming applications for real-time data processing and analytics.
- Hands on experience in working wif multiple tools in teh Hadoop ecosystem like SparkSQL, Spark Streaming, Hive, Pig, Sqoop, Kafka, MapReduce on various Hadoop distributions like Horton Works, Cloudera, EMR.
- Experienced in scheduling and workflow management using IBM Tivoli, Control-M, Oozie, and Airflow for efficient job orchestration.
- Analyze, design, build & deploy modern data solutions using Microsoft Azure PaaS / SaaS service to support visualization of data. Understand current Production state of application and determine the impact of new implementation on existing business processes.
- Strong database development skills in Teradata, Oracle, SQL Server, including the development of stored procedures, triggers, and cursors.
- Defined the digital technology Strategy, MarTech Roadmap and Platform Architecture to support omni-channel personalization, marketing attribution and account-based marketing (ABM)
- Proficient in version control systems like Git, GitLab, and VSS for code repository management and collaboration.

Education:

- Bachelors in Computer Science from JUNTU-Hyderabad, India (Aug 2008 – May 2012).
- Masters in Computer Science from Gannon university, Pennsylvania (Aug 2012- Dec 2013).

Technical Skills:

AWS Services	AWS s3 , redshift, EMR, SNS, SQS, aethna, glue, Terraform, cloudwatch, kenisis, route53, IAM
Big Data Technologies	HDFS, SQOOP, PySpark, hive, Greenplum, MapReduce, spark, spark streaming, Informatica,HBASE
Hadoop Distribution	Cloudera, Horton Works
Languages	Java, SQL, PL/SQL, Python, Unix, HiveQL, Scala.
Operating Systems	Windows (XP/7/8/10), UNIX, LINUX, UBUNTU, CENTOS.
Database	Teradata, oracle, SQL server, PostgreSQL.
Scheduling	IBM Tivoli,control-m,oozie,airflow
Version Control	GIT, GitHub, VSS
Methodology	Agile, Scrum.

IDE & BI Tools	Eclipse, Visual Studio, Tableau, PowerBI.
Other Cloud services	GCP, Big Query, Azure, Snowflake, Databricks.

Professional Experience:

Client: Safelite, Columbus, OH
Role: Sr. Data Engineer Dec 2023 – Till Now
Responsibilities:

- Designed and optimized Snowflake data pipelines to efficiently ingest and process high-volume semiconductor manufacturing data, utilizing Snowflake stages, transient, temporary, and persistent tables for structured storage.
- Developed ETL workflows using AWS Glue and AWS EMR, transforming raw semiconductor sensor data stored in AWS S3, and loading it into Amazon Redshift for yield analysis and predictive maintenance.
- Implemented Snowpipe for real-time data ingestion and AWS Kinesis for real-time data streaming, processing IoT and machine-generated data from semiconductor fabrication units for defect detection.
- Performed on Pentaho, Informatica, Talend Data Fabric ETL, Data warehouse concepts, Kimball, Star & Snowflake Schema, Fact/Dimension tables. Worked on Pentaho, Informatica, Talend/Informatica Integration Cloud Services, IaaS, PaaS, SaaS.
- Responsible for design and creation of Hive tables, partitioning, bucketing, loading data and writing hive queries. Implement ed, migrated existing Hive Script in SparkSQL for better performance.
- Developed big data processing pipelines using Hadoop ecosystem tools, including HDFS, Sqoop, Hive, MapReduce, Apache Spark, and PySpark, enabling large-scale analytics on semiconductor test and process control data.
- Designed and maintained OBIEE dashboards and reports for advanced data visualization and business intelligence, integrating data from Snowflake, Oracle, and Teradata to support supply chain and manufacturing analytics.
- Developed and optimized OBIEE RPD models, creating complex metadata layers to enhance data modeling and report generation for semiconductor production insights.
- Strong experience working with Spark (Dataframes and Spark Sql) for high performance data processing and data preparation.
- Deployed ETL mappings and BO Dashboard reports into production
- Proficient in data analysis, data validation, data cleansing, data verification and entire data science project life cycle.
- Integrated AWS SNS and SQS for event-driven messaging, ensuring real-time alerts for anomalies in semiconductor wafer production, and configured AWS Route53 for efficient network routing.
- Implemented AWS Athena for ad-hoc data querying on large-scale semiconductor datasets stored in AWS S3, and utilized AWS CloudWatch for monitoring, logging, and performance optimization of production systems.
- Designed and managed database solutions using Teradata, Oracle, MySQL, and SQL Server, optimizing structured data storage and retrieval for semiconductor manufacturing analytics.
- Worked with Systems Development Life Cycle (SDLC)/ Software as a Service (SaaS) delivery models
- Developed and scheduled workflows using IBM Tivoli, Control-M, Apache Oozie, and Apache Airflow, ensuring seamless execution of ETL jobs across Micron's global manufacturing facilities.
- Worked with Cloudera Hadoop distribution, utilizing Kafka and Zookeeper for real-time data ingestion and processing, and Ambari for cluster monitoring and management.
- Continuously update my knowledge and skills in AI/ML, IoT and Data Science
- Managed security and access control using AWS IAM, ensuring role-based access to cloud resources, and utilized JIRA for issue tracking and agile project management in semiconductor data engineering projects.
- Worked with Azure Data Factory (ADF) since its a great **SaaS solution** to compose and orchestrate Azure data services.
- Implemented Power BI DAX calculations and Power Query transformations to enhance data analysis and reporting accuracy, supporting regulatory compliance and business decision-making.

Environment: AWS, AWS S3, redshift, EMR, SNS, SQS, aethna, glue, cloudwatch, kenisis, route53, IAM, Sqoop, MYSQL, HDFS, Apache Spark, Hive, Cloudera, OBIEE, Kafka, Zookeeper, Oozie, PySpark, Ambari, JIRA, IBM Tivoli, control-m, OOZIE, airflow, Teradata, oracle, PowerBI, SQL.

Client: GE HealthCare, Milwaukee, Wi
Role: Data Analyst/ Engineer Feb 2022 – Nov 2023

Responsibilities:

- Designed and managed Snowflake tables (transient, temporary, persistent) to efficiently store and process large-scale healthcare claims and patient data while ensuring compliance with HIPAA regulations.
- Helped and trained e-commerce website admins to fully utilize admin panel, mainly for product category management, stock management and new product creation
- Optimized Snowflake performance by implementing advanced partitioning techniques and Snowflake caching mechanisms, accelerating medical data retrieval for real-time analytics.
- Hands-on using various AWS Services including EC2, EMR cluster, Redshift, Data Bricks, S3 Buckets, AWS Kinesis and IaaS /PaaS/SaaS.
- Conducted data blending, data deduplication, and data preparation using Alteryx, SQL, and DBT for Tableau consumption, automating transformation processes, and publishing data sources to Tableau Server.
- Developed and automated ETL workflows using AWS Glue and AWS Redshift, transforming raw patient records and insurance claims data into structured datasets for analytics and reporting.
- Done course work related to computer science: Data Structures, Application of Neural Network, Data Base Management Systems.
- Built end to end process for quote data analysis for Auto and Property product, to calculate win rates, bridge rates, quote rates month over month.
- Configured and fine-tuned Amazon Redshift clusters to handle high-volume healthcare data processing, optimizing query performance for policy risk assessments and member analytics.
- Integrated AWS SNS and SQS for real-time messaging, enabling proactive notifications for claim approvals, fraud detection, and policy renewals.
- Hands on experience in working on Spark SQL queries, Data frames, and import data from Data sources, perform transformations, perform read/write operations, save the results to output directory into HDFS.
- Implemented AWS Athena for ad-hoc querying on massive datasets stored in AWS S3, facilitating quick analytics on healthcare provider performance and patient outcomes.
- Developed data streaming pipelines using AWS Kinesis to process real-time insurance transactions, fraud detection alerts, and patient record updates.
- Managed DNS configurations and routing with AWS Route53, ensuring secure and seamless deployment of health insurance applications across cloud environments.
- Worked on Google Analytics data of e-commerce websites to prepare 270+ web/product analytics reports; enabled clients to improve their website contents and product specs.
- Built and optimized big data processing pipelines using Hadoop ecosystem tools (HDFS, Sqoop, Hive, MapReduce, Apache Spark, and Spark Streaming), supporting large-scale medical claims processing.
- Scheduled and automated workflows using IBM Tivoli, Control-M, Oozie, and Apache Airflow, ensuring efficient execution of ETL jobs and compliance reporting.
- Utilized open source data science language like python to develop novel tools and visualization approaches for data analysis and interpretation.
- Experience in Captiva Input Accel 6.5, 7.1 and 7.5 technologies by OpenText where all the customizations to the product are handled through .Net.
- Designed and managed relational databases in Teradata, Oracle, and SQL Server, optimizing data models for policy management, member eligibility, and provider networks.
- Integrated Quantexa's platform with existing data pipelines to enhance data enrichment, leveraging its capabilities in risk analysis and compliance reporting for healthcare providers and insurance companies.
- Proficient in utilizing version control systems such as Git, GitLab, and VSS for efficient code repository management and collaborative development processes.

Environment: AWS, AWS S3, redshift, EMR, SNS, SQS, Athena, glue, cloudwatch, kenisis, route53, IAM, Sqoop, MYSQL, DBT, HDFS, Apache Spark, Hive, Cloudera, Quantexa, Kafka, Zookeeper, Oozie, PySpark, Ambari, JIRA, IBM Tivoli, control-m, OOZIE, airflow, Teradata, oracle, SQL, Alteryx, Tableau.

Client: American Express, Phoenix, AZ

Role: Big Data Developer Oct 2020– Jan 2022

Responsibilities:

- Developed and managed data ingestion pipelines using Sqoop to import financial transaction data from MySQL (RDBMS) to HDFS, ensuring efficient data storage and retrieval.
- Performed large-scale data aggregations using Apache Spark and Scala, optimizing risk modeling and fraud detection algorithms, and storing results in Hive for further analysis.
- Worked extensively with Data Lakes and big data ecosystems (Hadoop, Spark, Hortonworks, Cloudera, Ambari) to enable real-time insights on customer spending patterns and credit risk assessments.
- Developed Hive queries and Spark SQL scripts to analyze structured and unstructured financial data, supporting compliance reporting and business intelligence.
- Integrated Kafka and Spark Streaming to process real-time credit card transactions, enhancing fraud detection and anomaly detection systems.
- Collaborated effectively with product and program managers to ensure project deliverables remained on track and aligned with business goals and expectations.
- Developed and scheduled Oozie workflows for batch processing of transaction data, automating ETL pipelines for financial risk analysis.
- Developed custom scripts and tools using Oracle's PL/SQL language to automate data validation, cleansing, and transformation processes.
- Migrated high-volume financial data from Oracle (RDBMS) to Hadoop using Sqoop, ensuring scalability and performance for machine learning-driven credit scoring models.
- Implemented PySpark and Spark SQL to optimize performance in fraud detection analytics and real-time customer behavior analysis.
- Utilized Zookeeper for cluster coordination and managed Spark Streaming to process continuous financial data feeds with minimal latency.
- Leveraged Git and JIRA for version control and issue tracking, ensuring seamless CI/CD integration and efficient project workflow management.
- Designed and developed ETL workflows using Informatica PowerCenter to extract, transform, and load large volumes of financial transaction data, optimizing data integration and ensuring data quality across enterprise data warehouses.
- Built Product report card Data Mart for Confidential, USA, allowing business users to generate product life cycle reports.
- Designed and developed Power BI dashboards and reports to provide interactive visualizations, KPIs, and real-time insights into financial transaction trends, fraud detection, and risk management.
- Explored with the Spark improving the performance and optimization of the existing algorithms in Hadoop using Spark Context, Spark SQL, Data Frame, Pair RDD and Spark on YARN.
- Worked with OBIEE for developing reports and dashboards, integrating data from multiple sources, and enhancing business intelligence capabilities for financial transaction analysis and compliance reporting.

Environment: Sqoop, MYSQL, HDFS, OBIEE, Apache Spark Scala, Hive Hadoop, Cloudera, Kafka, MapReduce, Zookeeper, Oozie, Data Pipelines, RDBMS, Python, PySpark, Ambari, Informatica, PowerBI, JIRA.

Client: MISI COMPANY, Bala Cynwyd, PA

Role: Hadoop Developer May 2018 – Sep 2020

Responsibilities

- Developed ETL jobs using Spark-Scala to migrate data from Oracle to new MySQL tables.
- Rigorously used Spark-Scala (RRD’s, Data frames, Spark SQL) and Spark- Cassandra-Connector API's for various tasks (Data migration, Business report generation etc.)
- Developed Spark Streaming application for real time sales analytics
- Prepared an ETL framework with the help of sqoop, pig and hive to be able to frequently bring in data from the source and make it available for consumption
- Processed HDFS data and created external tables using Hive and developed scripts to ingest and repair tables that can be reused across the project.
- Analyzed the source data and handled efficiently by modifying the data types. Used excel sheet, flat files, CSV files to generated PowerBI ad-hoc reports

- Analyzed the SQL scripts and designed the solution to implement using PySpark
- Extracted the data from other data sources into HDFS using Sqoop
- Handled importing of data from various data sources, performed transformations using Hive, MapReduce, loaded data into HDFS.
- Extracted the data from MySQL into HDFS using Sqoop
- Implemented automation for deployments by using YAML scripts for massive builds and releases
- Apache Hive, Apache Pig, HBase, Apache Spark, Zookeeper, Flume, Kafka and Sqoop.
- Implemented Data classification algorithms using MapReduce design patterns.
- Extensively worked on creating combiners, Partitioning, distributed cache to improve the performance of MapReduce jobs.
- Worked on GIT to maintain source code in Git and GitHub repositories

Environment: Hadoop, Hive, spark, PySpark, Sqoop, Spark SQL, Cassandra, YAML, ETL.

Client: GlaxoSmithKline, New Jersey, USA
Role: Data Warehouse Developer **Apr 2016 – May 2018**
Responsibilities:

- Creating jobs, SQL Mail Agent, alerts, and scheduling DTS/SSIS packages for automated processes.
- Managing and updating Erwin models for logical/physical data modeling of Consolidated Data Store (CDS), Actuarial Data Mart (ADM), and Reference DB to meet user requirements.
- Builds large-scale data processing systems in data warehousing solutions and works with unstructured data mining on NoSQL.
- Utilizing TFS for source controlling and tracking environment-specific script deployments.
- Exporting current data models from Erwin to PDF format and publishing them on SharePoint for user access.
- Developing, administering, and managing databases such as Consolidated Data Store, Reference Database, and Actuarial Data Mart.
- Writing triggers, stored procedures, and functions using Transact-SQL (T-SQL) and maintaining physical database structures.
- Deploying scripts in different environments based on Configuration Management and Playbook requirements.
- Creating and managing files and file groups, establishing table/index associations, and optimizing query and performance tuning.
- Tracking and closing defects using Quality Center for effective issue management.
- Maintaining users, roles, and permissions within the SQL Server environment.

Environment: SQL Server 2008/2012 Enterprise Edition, SSRS, SSIS, T-SQL, Windows Server 2003, Performance Point Server 2007, Oracle 10g, visual Studio 2010, Nosql.

Client: Johnson Controls, Orlando, FL
Role: Data Warehouse Developer **Feb 2014 – Mar 2016**
Responsibilities:

- Developed complex stored procedures, efficient triggers, and necessary functions, along with creating indexes and indexed views to optimize performance in SQL Server.
- Extensive experience in monitoring and tuning SQL Server performance, employing best practices to ensure optimal database performance.
- Expertise in designing ETL data flows using SSIS, creating mappings and workflows for extracting data from SQL Server, as well as performing data migration and transformation from Access/Excel sheets using SQL Server SSIS.
- Proficient in dimensional data modelling for Data Mart design, identifying facts and dimensions, and developing fact tables and dimension tables using Slowly Changing Dimensions (SCD) techniques.
- Skilled in error and event handling techniques, such as precedence constraints, breakpoints, check points, and logging, ensuring reliable and robust ETL processes.
- Experienced in building cubes and dimensions with different architectures and data sources for business intelligence purposes, including writing MDX scripting.
- Proficient in developing SSAS cubes, implementing aggregations, defining KPIs (Key Performance Indicators), managing measures, partitioning cubes, and creating data mining models. Deploying and processing SSAS objects.

- Experience in creating ad hoc reports and reports with complex formulas, utilizing querying capabilities of the database for business intelligence purposes.
- Expertise in developing parameterized, chart, graph, linked, dashboard, scorecards, and drill-down/drill-through reports on SSAS cubes using SSRS (SQL Server Reporting Services).

Education Details:

- MS SQL Server 2016, Visual Studio 2017/2019, SSIS, Share point, MS Access, Team Foundation server, Git.